

Simulation and representation of topology-changing rolling waves for massive open ocean games.

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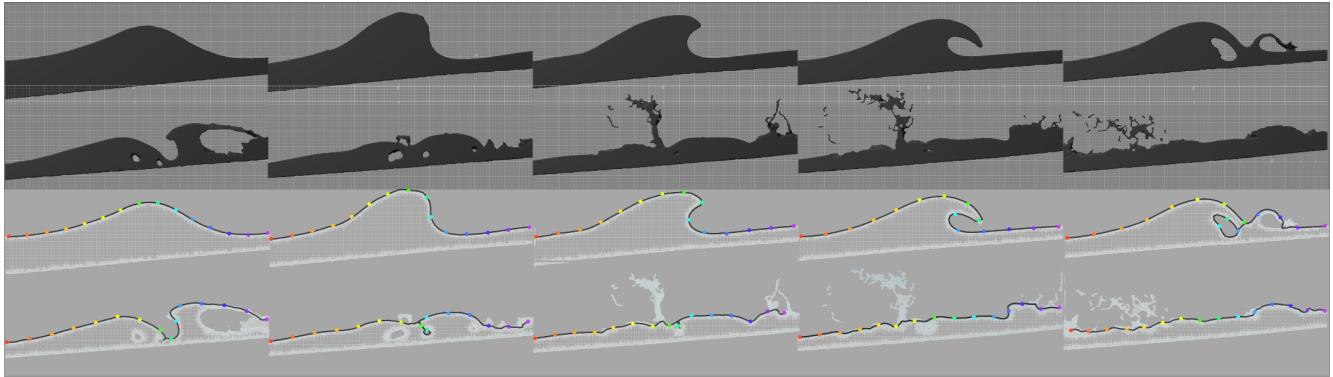


Figure 1: Rolling wave simulation surface (above) and the corresponding results of capturing the simulated surface with a curve (below).

ABSTRACT

Rendering close-shore water oceanic phenomena, e.g. rolling waves, for real-time applications is a challenging problem, especially in a massive open-world game where variability, artistic control and performance are crucial. While using a form of animated height displacement maps is a common technique in open-world games (e.g. *Assassin's Creed Odyssey* (2018), *Death Stranding* (2019)), it is limited to height data and is unable to deal with overhangs and changing topology of collapsing oceanic waves, we are proposing a novel method of capturing wave simulation data with a set of approximation curves that can deal with overhangs, changing topology and strict budget requirements.

KEYWORDS

simulation, rendering

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1 RENDERING OCEANIC PHENOMENA IN REALTIME

During the production of *Horizon Forbidden West*, we had to deal with the problem of rendering various oceanic effects in real-time, such as splashes and rolling waves in the setting of the massively open-world game. We encountered several challenges, such as simulating real-life oceanic phenomena and translating them to game content in a form suitable for artistic authoring post-simulation inside the game editor environment and real-time rendering in the game engine.

1.1 Changing topology

Because of the dynamic, chaotic nature of processes occurring in oceanic water waves collapsing, the resulting topology is unpredictable; in particular, it is suffering from multiple holes emerging in the process, thus making it conform to a topologically flat 2D surface that is also spatially and temporally coherent, particularly challenging. Initial inspiration comes from Hokusai (Figure 2), whose artistic observations lead us to leverage the fact that a rolling wave can be presented as an evolution of wave "profiles" spatially and temporally offset relative to each other and that various rolling waves are self-similar, hence isomorphic to a single archetypical wave with extra spatial and temporal deformations.

1.2 Capturing the simulation surface with curves

Such "profiles" must conform to the simulated surface, ignore high-frequency detail, and deal with topological holes from air pockets



Figure 2: The Great Wave off Kanagawa. Print at the Metropolitan Museum of Art

after the wave collapses. We begin with a topologically flat simulation frame and project the curve onto the surface. In the next frame, we reproject the curve to the closest surface; we advect the curve points with velocities from simulation data and resample the curve, keeping the number of vertexes consistent. An additional entanglement check removes points that may be advected ahead of closest neighbors. The following formula captures the iterative nature of reprojection, advection, and resampling steps:

$$\mathbf{P}_{(t)} = \sum (\text{project} \cdot \text{advection} \cdot \text{resample}) \mathbf{P}_{(t-1)} dt$$

Which results in a profile-curve construction, as shown in Figure 1.

1.3 Filtering the curve data

The resulting set of curves per frame generates uv-coordinates, where each point uv coordinate is mapped in the uv-square formed by normalized curve length as v and normalized time-frame dimension as u . The curve length also changes because of the changing curvature, leading to noticeable jitters of texture coordinates unless it is temporally filtered. Hence, each curve uv-space must be adjusted and smoothed relative to the neighbors.

1.4 Capturing profile curves as texture maps

When the wave-profile curves are generated and temporally smooth-filtered, the resulting uv-space can be used to capture necessary data for material expression, e.g., color and position. Effectively, each texture column encodes a curve frame, which can be read left-to-right in a displacement material. 256 by 256 pixels, uncompressed, 16-bit integer format was used for the displacement map; 8-bit RGBA was used for the color map (see Figure 3(left,middle)).

1.5 Using 3D displacement maps in ocean rendering

After the color and position (displacement) data is captured as textures, it can be used in fragment and displacement shaders. We used an in-house technology dubbed “wave-particles” and time-curves to support rendering this data to displace the ocean surface in real time (Figure 3(right)). Artist-controlled time curves are exposed

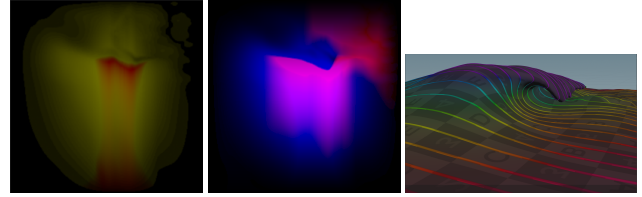


Figure 3: (left) displacement map, where red and green channels store normalized direction and displacement respectively; (middle) color map encoding foam trail and bubble data in red and blue channels; (right) result of wave reconstruction using wave-profile encoded textures

to artists as editable splines in our game editor, allowing control of individual wavefronts and timings of wave collapse.

2 LIMITATIONS

The method is known to lead to artifacts in situations when multiple “wave particles” overlap and “wave profiles” from different time-frames are conflicting. This is addressed by manual authoring of wavefronts positions and timing.

3 FUTURE WORK

The timing and dynamics of the wave collapse could be improved by introducing noisy time offset maps that add extra variability to wave behavior. The tessellation of the displaced surface can be further improved by resampling the wave curves that preserves sharp details. While the current method may be limited to effectively faking the 3D phenomena with a set of 2D slices + time, we have explored the possibility of generalizing it for capturing arbitrary ocean surface phenomena, such as splashes. We encountered significant problems leading to temporarily inconsistent results, but we plan to address that in future work.

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